

KINETIC TYPOGRAPHY SUBTITLE ENGINE - MASTER PROJECT BLUEPRINT

1. Project Overview & Vision

Project Name: Kinetic Typography Subtitle Engine (Internal Codename: The Lyric Engine) **Core Goal:** Ek aisa advanced subtitle generation tool banana jo boring, static text ko highly aesthetic aur animated typography mein convert kare. **Primary Output:** Transparent video files (.webm / .mov with alpha channel) ya Advanced Subtitle files (.ass) jinko manual video editors mein directly overlay kiya ja sake. **(MVP targets 9:16 Aspect Ratio for Shorts/Reels).** **Future Vision:** Is tool ko ek SaaS product mein convert karna aur ek YouTube channel ke through monetize/market karna.

2. Target Audience & Use Cases

- **Content Creators:** YouTube Shorts, Instagram Reels, aur TikTok creators.
- **Music Channels:** Lofi, Slowed + Reverb, Phonk, aur Pop music lyric videos.
- **Video Editors:** Jo manually subtitle animate karne mein ghanto waste karte hain.

3. System Architecture (The 3-Step Decoupled Hybrid Model)

System ko 3 independent modules mein divide kiya gaya hai:

Module 1: The Timing Extractor (Audio-to-Data)

- **Input:** User audio/video file upload karega.
- **Process:** Ek advanced Speech-to-Text model (**ElevenLabs Scribe v2**) audio process karega for extremely accurate word-level timestamps.
- **Output:** JSON/Text file jisme har word ka exact Start-Time aur End-Time (Timestamp) hoga.

Module 2: The AI Stylist (Auto-Tagging Agent)

- **Input:** Extracted Text aur gaane ka context/mood.
- **Process:** Ek fast LLM (Gemini Flash / GPT-4o-mini) text ko analyze karega aur automatically "Custom Tags" add karega. (Uses Chunking for long audio).
- **Output:** Tagged script (Zero manual hardwork).

Module 3: The Styling Renderer (The Core Engine)

- **Input:** Timing Data + Tagged Script + Selected Master Theme.
- **Process:** Engine code in inputs ko mila kar animations, colors, aur transitions apply karega. (Rendered in 9:16).
- **Output:** Final rendered video ya .ass subtitle file.



4. Core Features & Controls

A. Custom Tags (The Markup Language)

- **Basic:** `<big>` , `<glow:color>` , `<fill>` , `<font:name>` , `<space:wide>`
- **Advanced:** `<shake>` , `<glitch>` , `<texture:fire/water>` , `<echo>` , `<reveal:mask>` , `<kinetic:stretch>` , `<3d:flip>`

B. Master Themes (Global 1-Click Styles)

1. The Phonk / Aggressive
2. The Lofi / Vaporwave
3. The Pop / Dynamic Karaoke
4. The Minimalist Cinematic



5. Edge Cases & Risk Mitigation

1. **Aspect Ratio Constraint (9:16):** Ensuring the renderer outputs strictly vertical videos, requiring robust pagination and auto-wrap logic for lyrics.
2. **LLM Chunking:** Processing large subtitle files in smaller logical chunks to avoid LLM token limits and JSON hallucination.
3. **Text Overflow:** Engine mein "Max Words Per Line" aur "Auto-Wrap" logic.
4. **Timing Fallback:** Default duration fallbacks if API fails.
5. **The 'Tofu' Problem:** Fallback font system for emojis/special characters.



6. Tech Stack

- **AI Models:** ElevenLabs Scribe v2 (for audio STT), Gemini 1.5 Flash / GPT-4o-mini (for AI Stylist).
- **Backend/Scripting:** Python.
- **Rendering Engine:** MoviePy (for MVP).